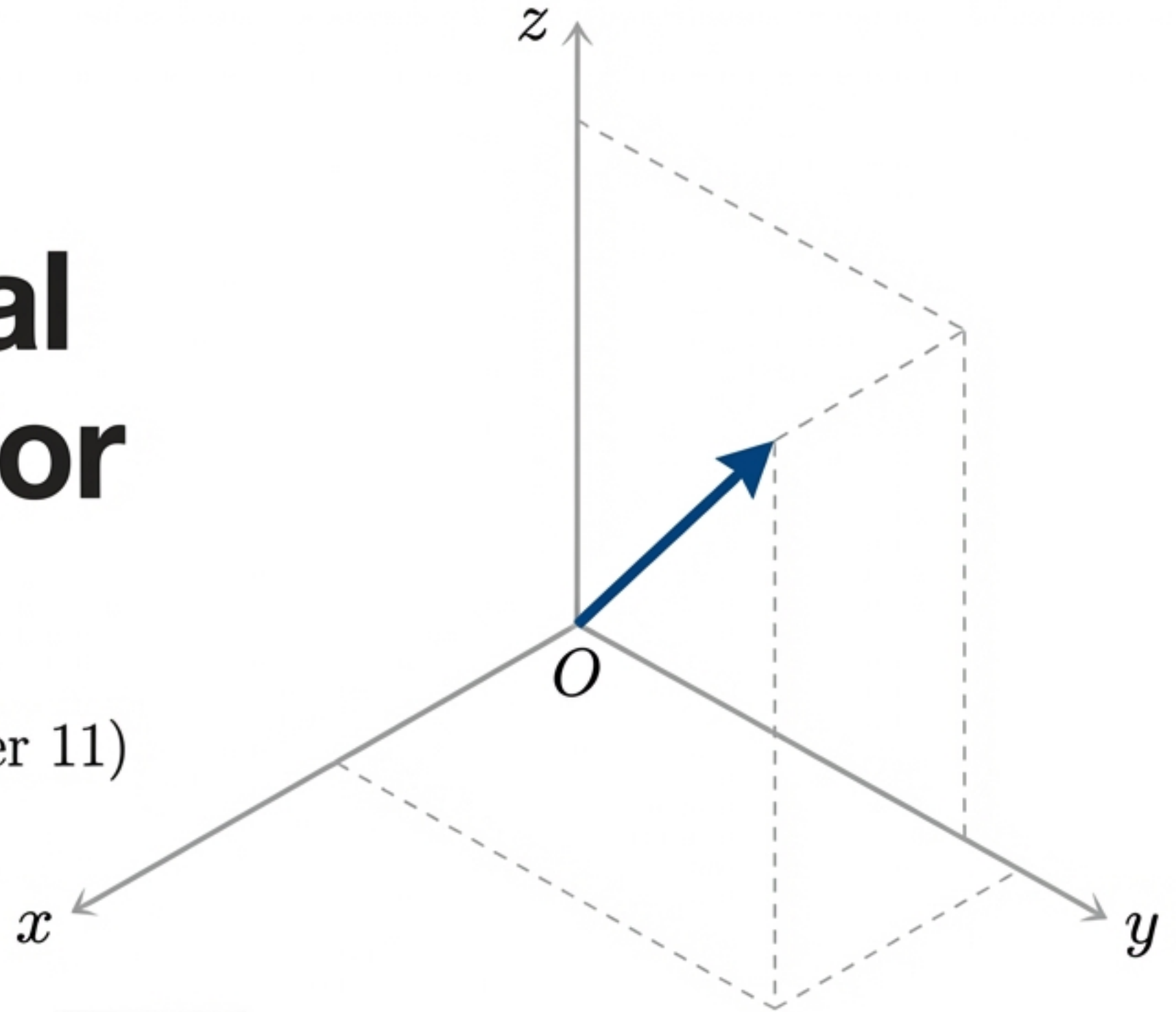


Three Dimensional Geometry: A Vector Approach

Distilling Class 12th Mathematics (Chapter 11)

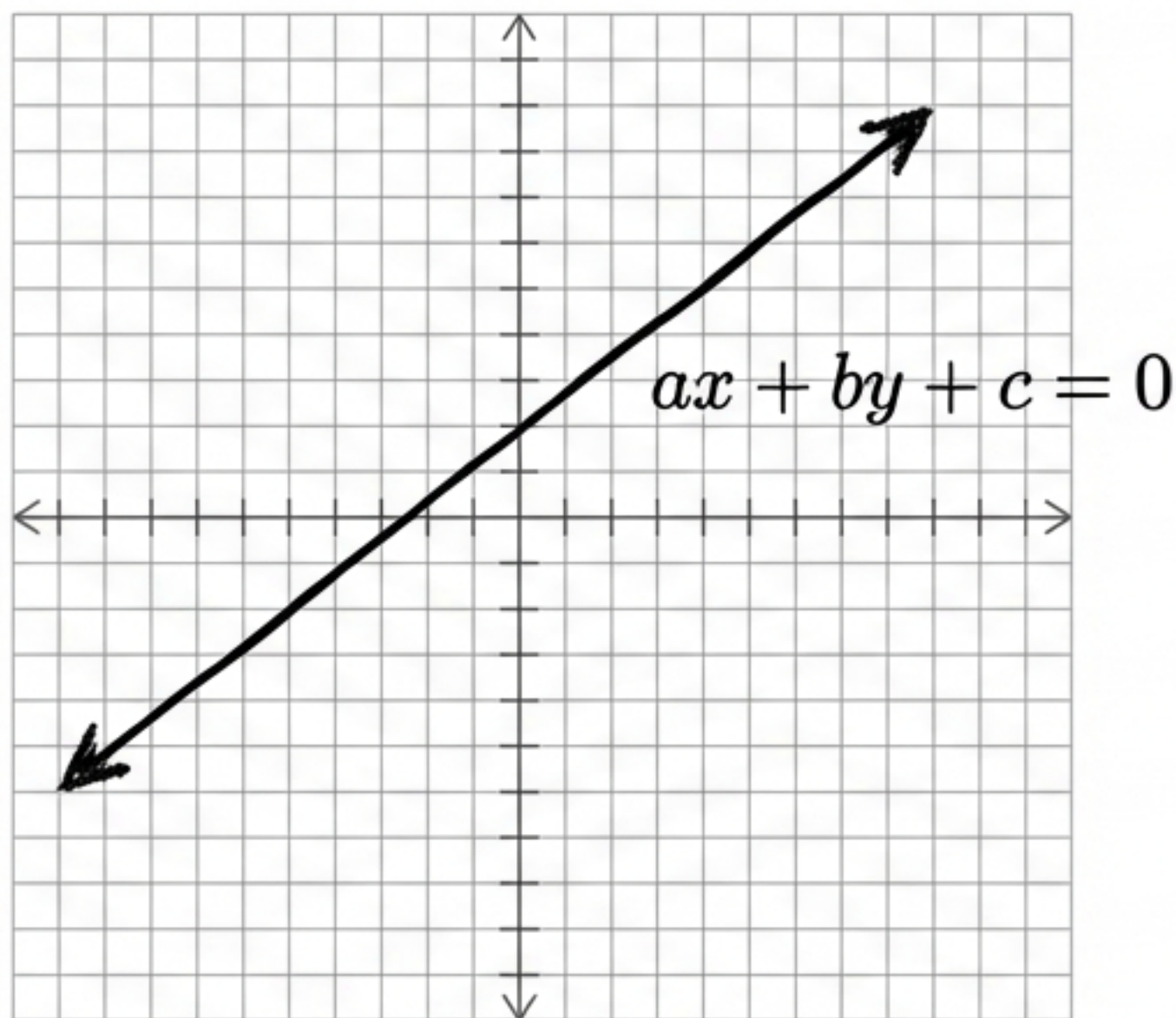


“The moving power of mathematical invention is not reasoning but imagination.” – A. De Morgan

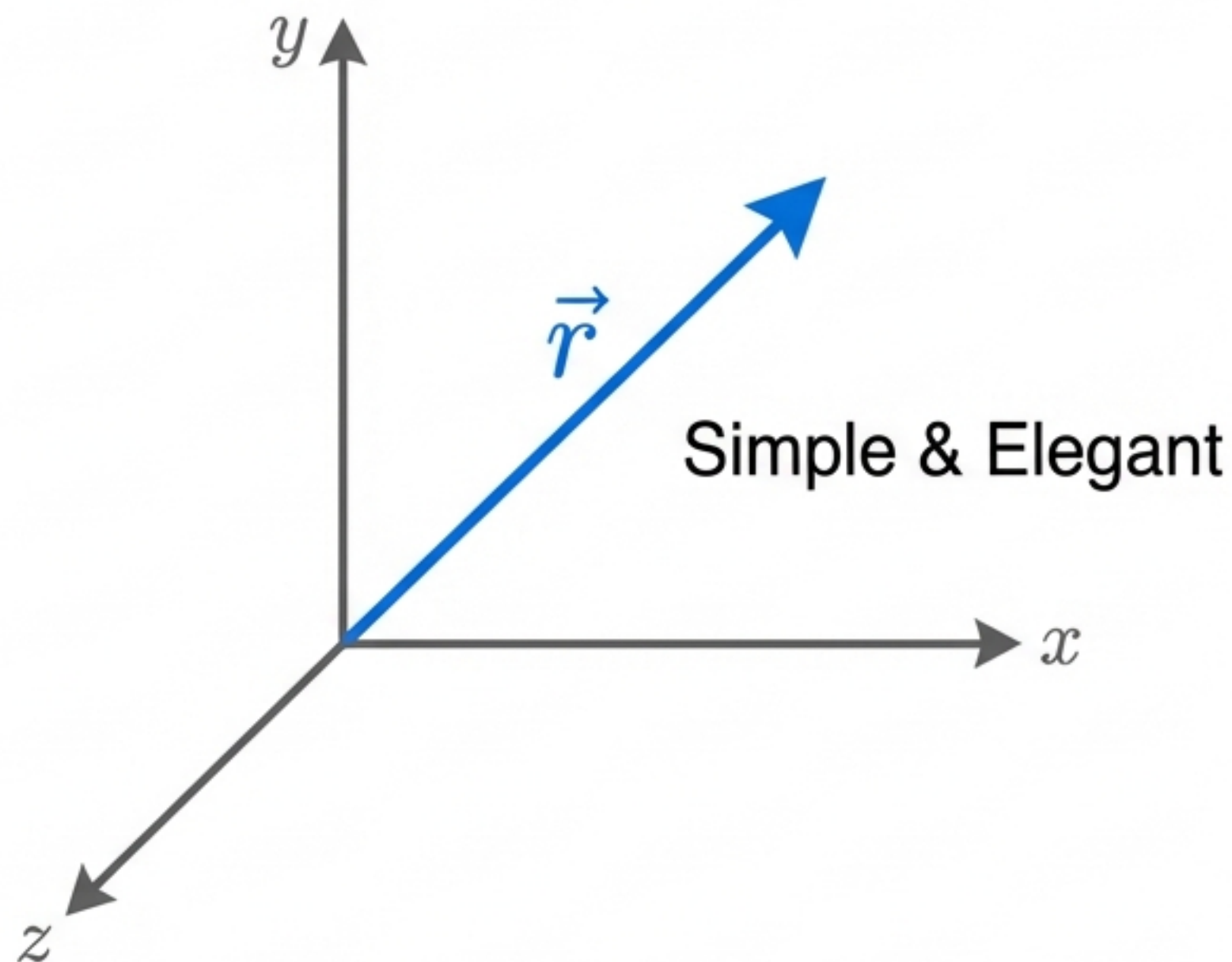


The Vector Advantage

The Past: Cartesian 2D

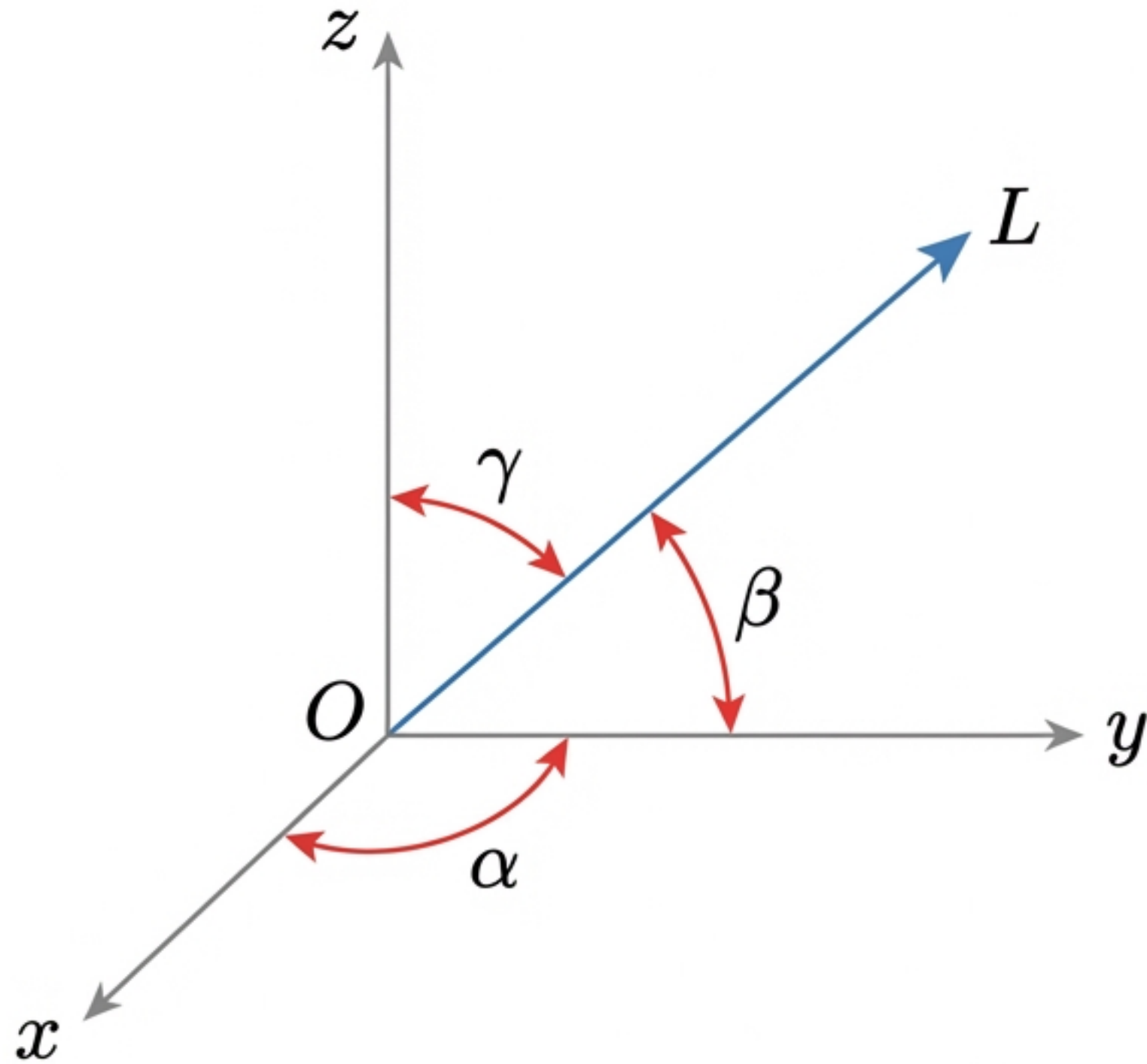


The Present: Vector 3D



We are moving from coordinate-heavy methods to Vector Algebra. This approach distills complex spatial relationships into compact, elegant equations.

The DNA of Direction: Direction Cosines (D.C.s)



Direction Angles:

$$\alpha, \beta, \gamma$$

Direction Cosines:

$$l = \cos \alpha$$

$$m = \cos \beta$$

$$n = \cos \gamma$$

The Golden Rule:

$$l^2 + m^2 + n^2 = 1$$

Direction Cosines are unique for any directed line.

Practical Direction: Direction Ratios (D.R.s)

Definition

Direction Cosines (l, m, n) are precise but often irrational numbers.

Direction Ratios (a, b, c) are any three numbers proportional to them.

$$a \propto l, \quad b \propto m, \quad c \propto n$$

The Relationship

Since they are proportional, we can write:

$$a = kl, \quad b = km, \quad c = kn$$

The Conversion Tool

How to recover Cosines from Ratios:

$$l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \quad m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \quad n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

Unlike Cosines, a line has infinitely many sets of $\vec{r}$

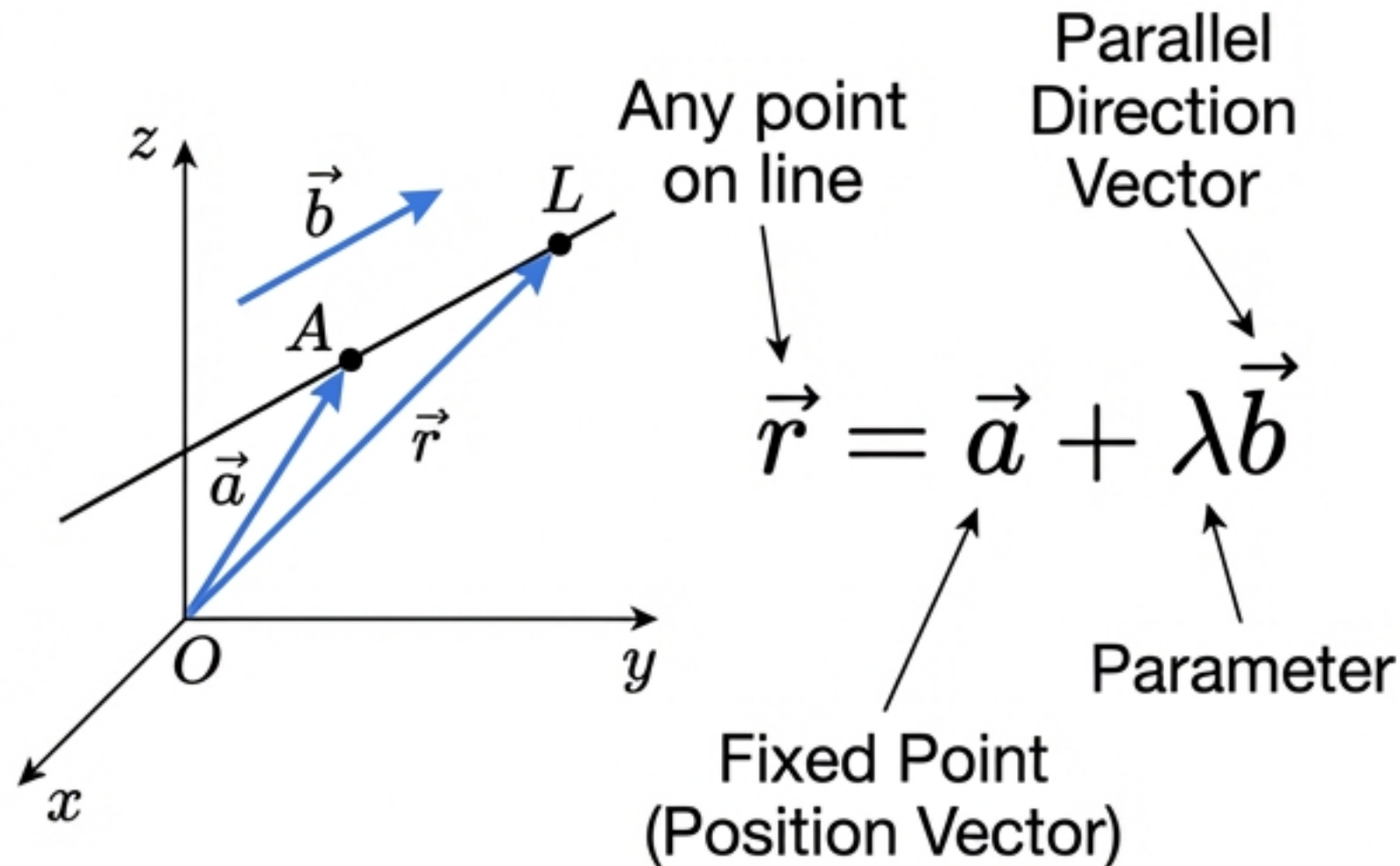


Direction Ratios.

Constructing a Line: Point & Direction

Given a fixed point A and a direction vector $\vec{b}$

VECTOR FORM



CARTESIAN FORM

Coordinates of Fixed Point

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

Direction Ratios

Constructing a Line: Through Two Points

Given points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$

VECTOR FORM

$$\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$$

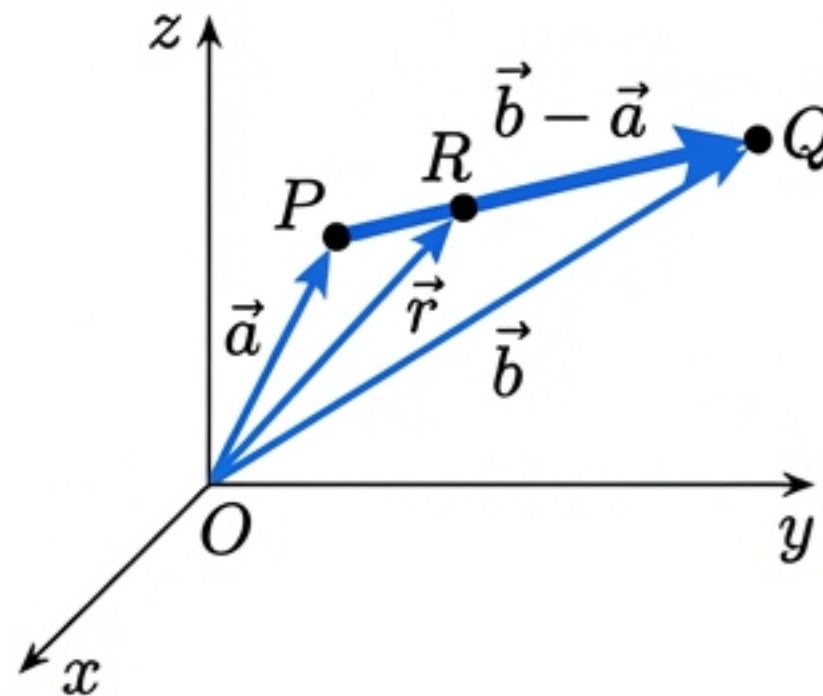
The direction is derived by subtracting the position vectors.

$$\text{Direction} = \vec{b} - \vec{a}$$

CARTESIAN FORM

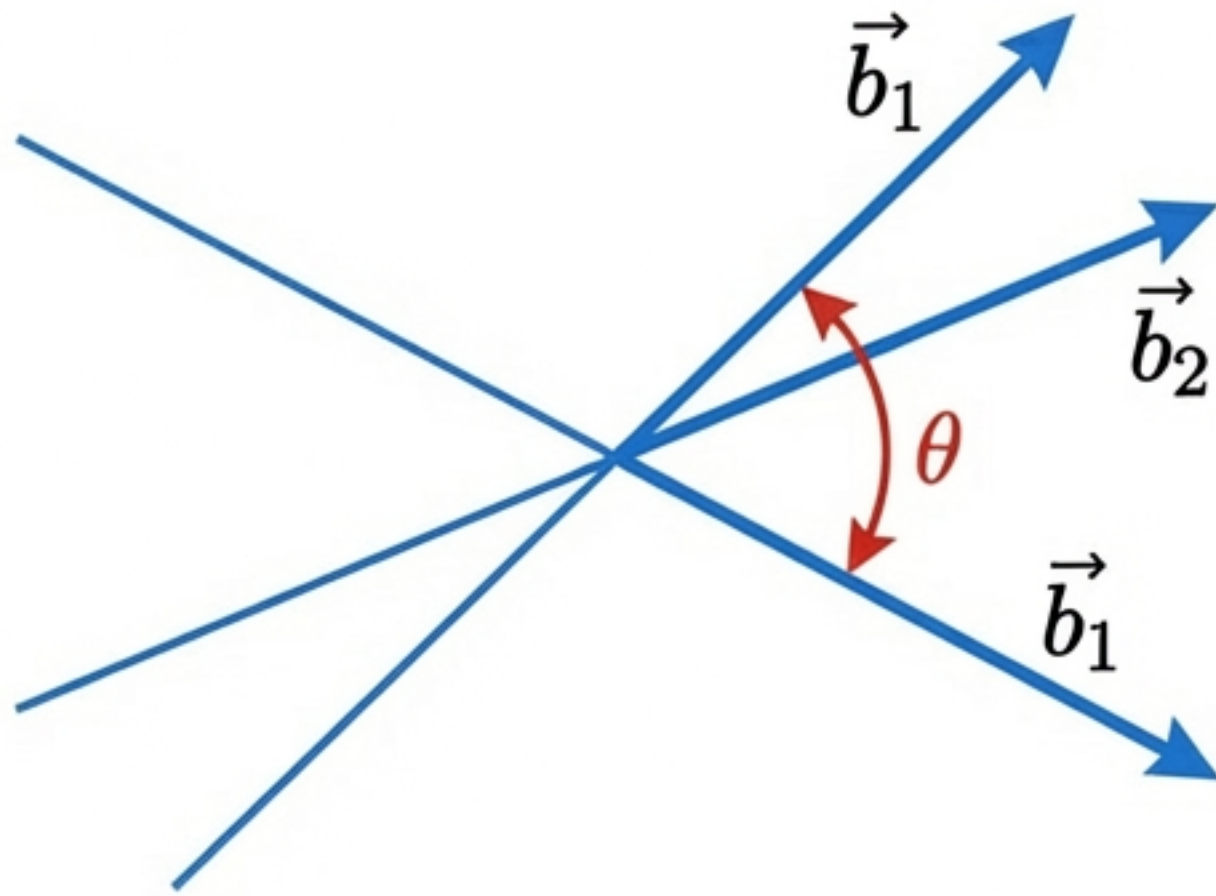
$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

The direction ratios are the differences in coordinates.



Interaction: The Angle Between Two Lines

The angle θ between two lines depends **only** on their direction vectors ($\vec{b}_1$ and $\vec{b}_2$). Their starting positions are irrelevant.



Vector Form (Primary):

$$\cos \theta = \left| \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1| |\vec{b}_2|} \right|$$

Cartesian Form:

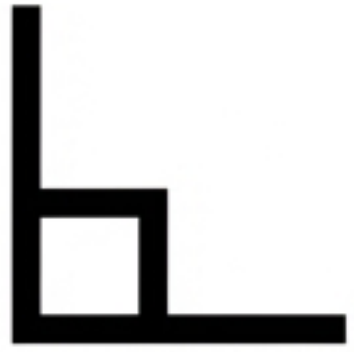
$$\cos \theta = \left| \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} \right|$$

Using Direction Cosines:

$$\cos \theta = |l_1 l_2 + m_1 m_2 + n_1 n_2|$$

Critical Conditions

PERPENDICULAR $\perp$



Angle $\theta = 90^\circ$. The Dot Product is zero.

Vector Condition: $\vec{b}_1 \cdot \vec{b}_2 = 0$

Cartesian Condition: $a_1a_2 + b_1b_2 + c_1c_2 = 0$

PARALLEL $\parallel$



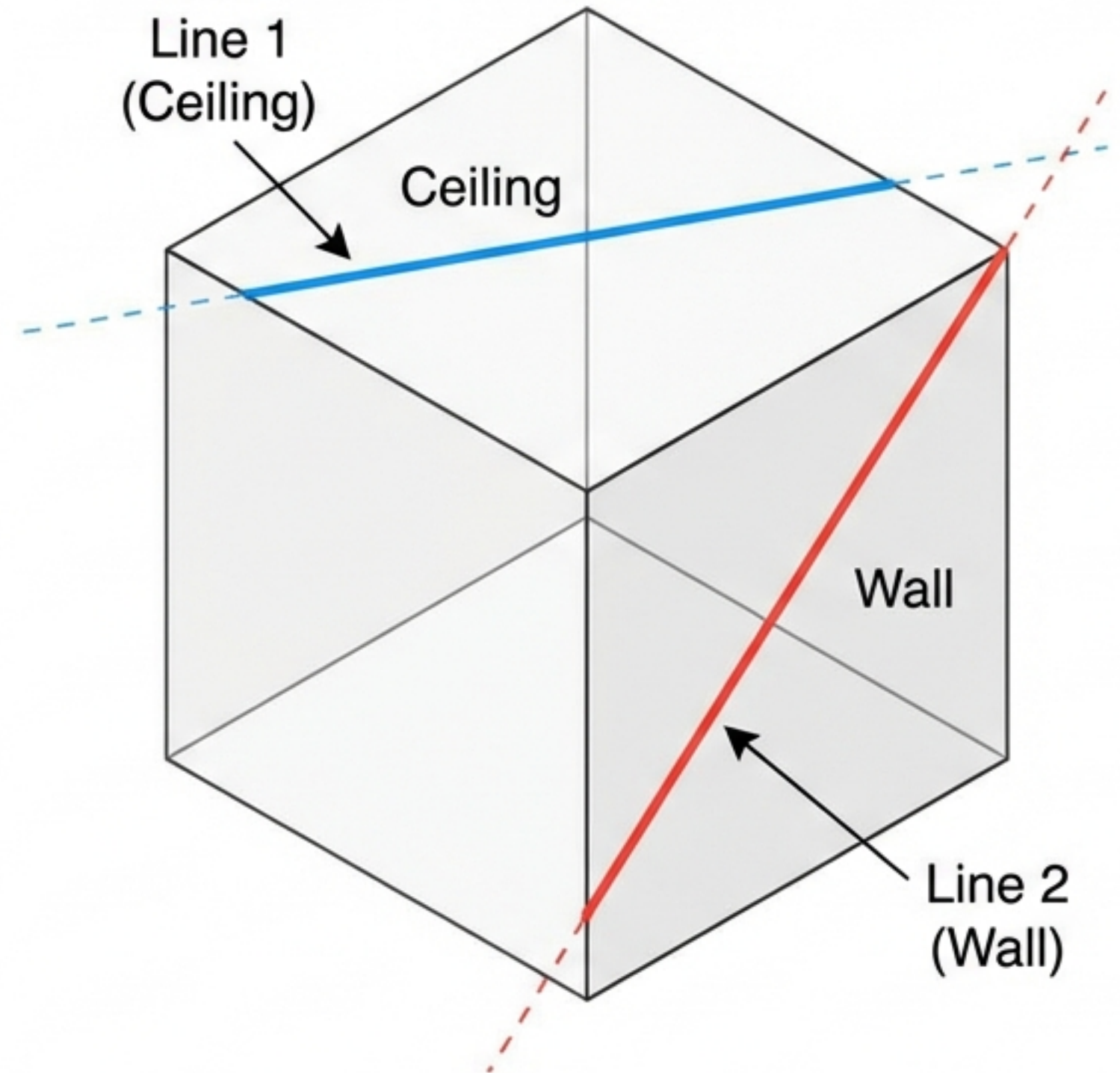
Angle $\theta = 0^\circ$. Direction Ratios are proportional.

Vector Condition: $\vec{b}_1 = \lambda \vec{b}_2$

Cartesian Condition: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

The Mystery of Skew Lines

Definition: In 3D space, lines do not have to intersect to be non-parallel. Skew Lines are lines that are neither parallel nor intersecting. They lie in different planes.



Measuring the Gap: Shortest Distance (Vector Form)

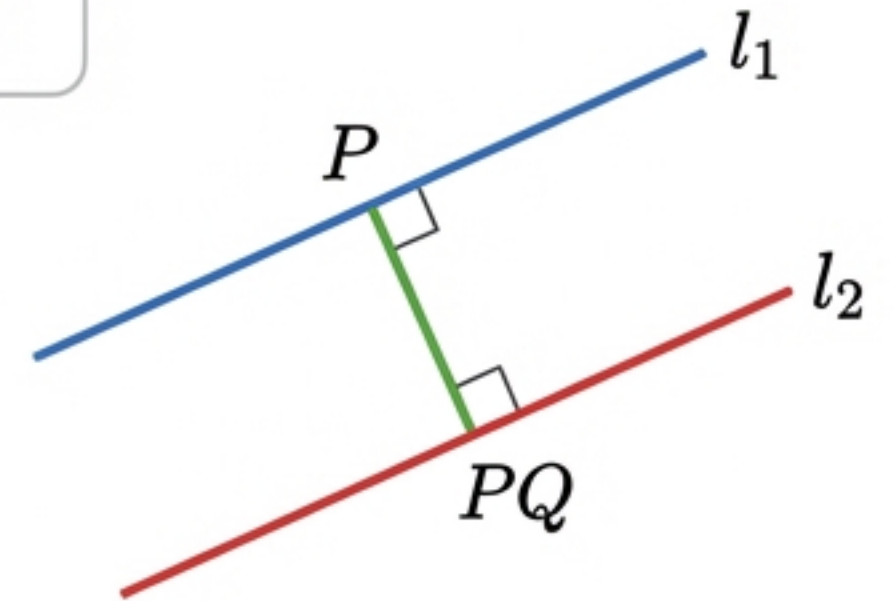
Finding the length of the segment perpendicular to both skew lines.

The unit vector
perpendicular to BOTH lines
(The Cross Product).

The vector connecting
the starting points.

$$d = \left| \frac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

Projecting the connection
onto the perpendicular.



Shortest Distance: Cartesian Form

The same calculation, expressed using coordinates. This effectively computes the volume of a parallelepiped and divides by the area of the base.

$$d = \frac{\text{Determinant}}{\sqrt{\dots}}$$

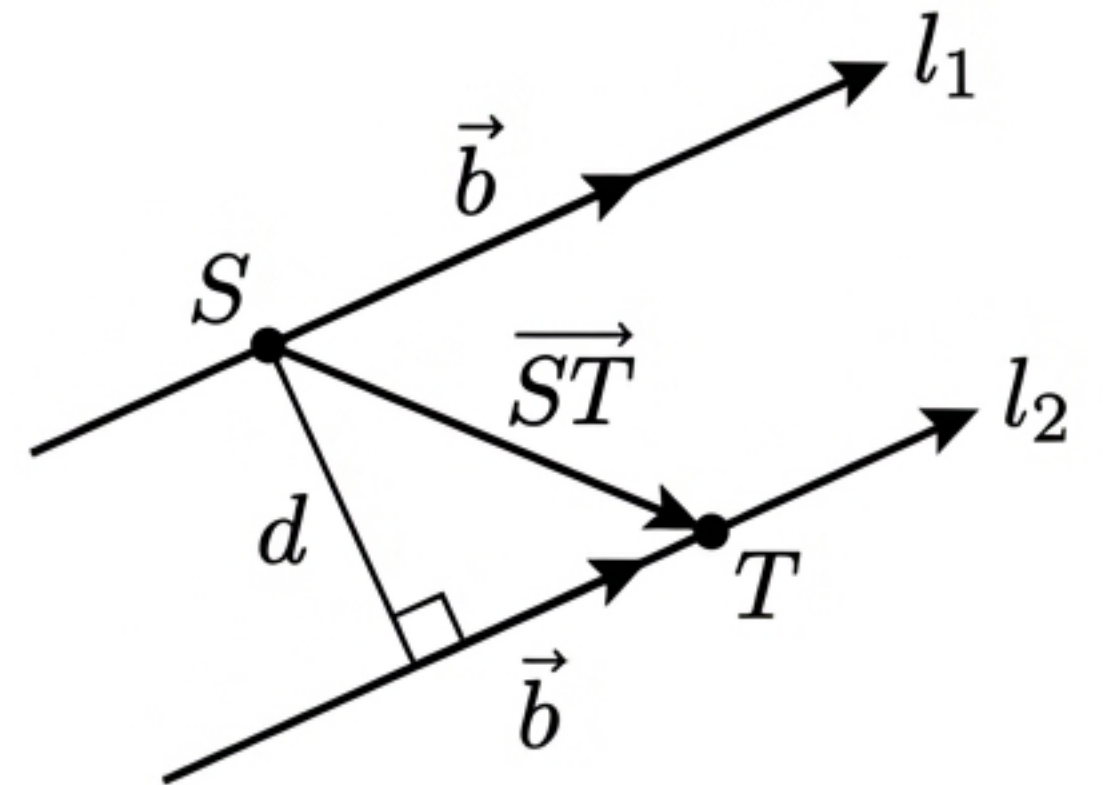
$$\text{Num} = \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$$

Legend: Black Row: Difference of Points ($x_2 - x_1, \dots$); Blue Row: Direction Ratios of Line 1; Red Row: Direction Ratios of Line 2

Special Case: Distance Between Parallel Lines

If lines are parallel, their direction vectors are the same ($\vec{b}$). The previous cross product formula ($\vec{b} \times \vec{b}$) would result in zero, so we use a specific variation.

$$d = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$$



In Practice: Calculating Shortest Distance

Problem: Find distance between:

Line 1: $\vec{r} = (\hat{i} + \hat{j}) + \lambda(2\hat{i} - \hat{j} + \hat{k})$

Line 2: $\vec{r} = (2\hat{i} + \hat{j} - \hat{k}) + \mu(3\hat{i} - 5\hat{j} + 2\hat{k})$

Solution Steps:

1. Identify: $\vec{a}_1 = \hat{i} + \hat{j}, \quad \vec{b}_1 = 2\hat{i} - \hat{j} + \hat{k}$
 $\vec{a}_2 = 2\hat{i} + \hat{j} - \hat{k}, \quad \vec{b}_2 = 3\hat{i} - 5\hat{j} + 2\hat{k}$

2. Difference: $(\vec{a}_2 - \vec{a}_1) = \hat{i} - \hat{k}$

3. Cross Product $(\vec{b}_1 \times \vec{b}_2) := 3\hat{i} - \hat{j} - 7\hat{k}$ (Magnitude $\sqrt{9 + 1 + 49} = \sqrt{59}$)

4. Dot Product: $(3\hat{i} - \hat{j} - 7\hat{k}) \cdot (\hat{i} - \hat{k}) = 3(1) + (-1)(0) + (-7)(-1) = 10$

****Result:** $d = \frac{10}{\sqrt{59}}$

Summary: The 3D Geometry Toolkit

DIRECTION

Cosines: l, m, n ($l^2 + m^2 + n^2 = 1$)

Ratios: a, b, c

Convert: $l = \pm a / \sqrt{a^2 + b^2 + c^2}$

ANGLES

Angle: $\cos\theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1||\vec{b}_2|}$

Perpendicular: $\vec{b}_1 \cdot \vec{b}_2 = 0$

Parallel: $\vec{b}_1 = \lambda \vec{b}_2$

LINE EQUATIONS

Vector: $\vec{r} = \vec{a} + \lambda \vec{b}$

Cartesian: $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$

2-Point: $\vec{b} \rightarrow (\vec{b} - \vec{a})$

DISTANCE

Skew: $d = \left| \frac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|} \right|$

Parallel: $d = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$

Conclusion

We have moved from the definition of a single direction to the measurement of invisible gaps between infinite skew lines.

By utilizing **Vector Algebra**, we transformed complex spatial geometry into elegant algebraic equations.

This framework forms the mathematical backbone of physical space, architecture, and computer graphics.